

REVIEW



An outlook on the trigeminovascular mechanisms of action and side effects concerns of some potential neuropeptidergic antimigraine therapies

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ABSTRACT

Introduction: In addition to serotonin (5-hydroxytryptamine; 5-HT) and other (neuro)mediators, the role of neuropeptides in migraine pathophysiology is relevant. Indeed, while some molecules interfering with calcitonin gene-related peptide (CGRP) transmission have recently been approved for clinical antimigraine use, other neuropeptides with translational use are in the pipeline. Among others, hypothalamic neuropeptides such as pituitary adenylate cyclase-activating peptide (PACAP), oxytocin (OT), and orexins stand out as potential novel targets to treat this neurovascular disorder.

Areas covered: Based on the aforementioned findings, the present review: (i) summarizes the current knowledge on the role of the above neuropeptides in the trigeminovascular system, and migraine pathophysiology; and (ii) discusses some issues related with the mechanisms of action and side effects concerns that could be elicited when targeting the CGRPergic, PACAPergic, oxytocinergic and orexinergic systems.

Expert opinion: Specific antimigraine pharmacotherapies have evolved from the enhancement of serotonergic 5-HT_{1B/1D/1F} transmission to the use of compounds interacting with neuropeptidergic systems. Canonically, neuropeptides cause an array of complex intracellular mechanisms that, after modifying neuronal and/or vascular transmission, result in antimigraine action and also potential side effects. Furthermore, due to the chemical nature of some molecules targeting the above neuropeptidergic transmission (e.g., monoclonal antibodies, peptides), there are some limiting pharmacokinetics issues.

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1. Introduction

1.1. Migraine as a neurovascular disorder

According to the International Headache Society (IHS), migraine is a common disabling primary headache, where a complex neurovascular cascade of events leads to a severe unilateral headache with nausea, vomiting, photophobia, phonophobia, and other neurological-associated symptoms [1]. Clinically, migraine consists of three phases: a trigger, an aura, and a headache phase. In general terms, this disorder can be classified into episodic migraine (EM) and chronic migraine (CM), and in both cases, it comprises two main subtypes (see Table 1 for details), namely: (i) migraine without aura (characterized by a headache with precise features and associated symptoms); and (ii) migraine with aura (in which the headache is preceded by transitory neurological symptoms including blurred vision, vertigo, numbness, tingling, etc.) [1]. Despite the many studies and progress made over the years, the reality is that the mechanisms involved in the pathophysiology of migraine are not yet fully characterized. Notwithstanding, as recently summarized [2–4], the numerous lines of evidence obtained from these studies suggest that

migraine is a pathophysiological state that may involve, among other changes:

- (i) Low plasma concentrations of serotonin (5-hydroxytryptamine; 5-HT) and/or high plasma concentrations of the calcitonin gene-related peptide (CGRP) which, in turn, favor the activation of the trigeminovascular system [5].
- (ii) Specific vasodilatation of intracranial and extracranial arteries, which are a significant source of pain in migraine, seemingly associated with CGRP release from trigeminal sensory nerves. This CGRP-induced vasodilatation, in turn, mechanically activates perivascular trigeminal sensory nerves resulting in peripheral and central trigeminal sensitization [6].

From the above findings, apart from activation of the serotonergic 5-HT_{1B/1D/1F} system, blocking the CGRPergic transmission is a major therapeutic area for treating migraine. Certainly, blocking the CGRPergic transmission is, at least partly, due to the reduction of CGRP release induced by activation of pre-junctional 5-HT_{1B/1D/1F} receptors on the trigeminovascular system (at peripheral and trigeminal level) [2–9]. These lines of

Article highlights

- Understanding the role of the trigeminovascular system in the pathophysiology of migraine has been relevant in the development of novel antimigraine pharmacotherapies.
- Currently, migraine is considered a neurovascular disorder associated with: (i) low plasma concentrations of serotonin (5-hydroxytryptamine; 5-HT) and/or high plasma concentrations of calcitonin gene-related peptide (CGRP); (ii) vasodilatation of intracranial and extracranial arteries, which are a significant source of pain in migraine, seemingly associated with CGRP release from trigeminal sensory nerves; and (iii) activation of the trigeminovascular system at the peripheral and central levels.
- Consequently, and regardless of their site of action, the development of molecules interfering with CGRPergic transmission (gepants and monoclonal antibodies [mAbs]) offers a considerable improvement to prevent migraine attacks.
- Despite this progress, not all patients respond to current pharmacotherapies; while the development of molecules with minor adverse events is necessary. This implies that, in some patients, several other molecules may also be involved. Within this context, various hypothalamic neuropeptides such as pituitary adenylate cyclase-activating peptide (PACAP), oxytocin (OT) and orexins also seem to play a role in migraine pathophysiology.
- These hypothalamic neuropeptides exert important effects in the trigeminovascular system by facilitating or disrupting the nociceptive trigeminal transmission.
- mAbs against PACAP are currently under randomized clinical trials.
- More recently, OT has emerged as an analgesic neuropeptide, and some small clinical assays suggest that intranasal delivery of this molecule results in antimigraine effects.
- Disrupted orexinergic systems can contribute to migraine nociception; as such, this system represents a potential therapeutic target for antimigraine action.
- This box summarizes key points contained in the article.

evidence have led to the development of effective and selective antimigraine drugs [7–9], including: (i) triptans (5-HT_{1B/1D/1F} receptor agonists; *e.g.*, sumatriptan); (ii) ditans (selective 5-HT_{1F} receptor agonists; *e.g.*, lasmiditan); (iii) gepants (CGRP receptor antagonists; *e.g.*, ubrogepant, rimegepant and atogepant); and (iv) monoclonal antibodies (mAbs) that directly block CGRP (*e.g.*, eptinezumab, fremanezumab and galcanezumab) or its receptor (*e.g.*, erenumab) [7–9].

Nevertheless, in addition to their potential to produce some adverse events, the above antimigraine pharmacotherapies are not effective in all patients. This may be due, at least in part, to the complex pathophysiology of migraine where, apart from CGRP, other endogenous molecules may also play a role (*e.g.*, hypothalamic neuropeptides [10]). Certainly, several systems seem to be deregulated before, during, and after a migraine attack; however, despite the availability of specific antimigraine drugs, not all patients respond to these treatments. For example, in the case of triptans (preferential 5-HT_{1B/1D/1F} receptor agonists), one-third of migraine patients do not respond to them; and apart from contraindications in patients with cardiovascular diseases, these acute antimigraine agents cannot be used as a prophylactic treatment since they can produce medication-overuse headache (MOH) [11]. In any case, it is interesting to note that the current specific drugs (*i.e.*, ergots, triptans, ditans, gepants and mAbs) to treat migraine not only disrupt

pain transmission (at the peripheral and/or central levels) but, in the case of the ergots and triptans, they can also elicit vascular (vasoconstrictor) effects.

The present review summarizes the current knowledge about some neuropeptidergic systems that are targeted by current or prospective pharmacotherapies interacting with the CGRPergic, PACAPergic, oxytocinergic and orexinergic systems.

1.2. The trigeminovascular system and its role in migraine pathophysiology

Migraine is more than a pain disorder, and the current consensus on its pathophysiology mainly relies on the neurovascular theory. Indeed, it is considered that headache pain is partly due to the vasodilatation of extracranial and intracranial blood vessels, which in turn produces mechanical activation of perivascular trigeminal fibers and sensitization of central terminals and second-order neurons of the trigeminocervical complex [12]. Beyond the controversies about the origin of pain [13,14], the migraine trigger has not been elucidated and it seems to be multifactorial (environmental and genetic factors) [15]. Admittedly, the exact origin of the trigeminovascular pathways causing a migraine attack remains elusive, but it has been proposed [5,16–19] that: (i) at the peripheral level, the activation and sensitization of these pathways results in CGRP release from trigeminal perivascular nerves and, consequently, in peripheral (including intra/extracranial) vasodilatation; and (ii) at the central level, trigeminal CGRP release may facilitate the trigeminal nociceptive transmission [5,16–19].

Furthermore, although we are trying to explain the pathophysiology of migraine in terms of a dysfunction in the trigeminovascular system, subcortical and cortical structures influence (a top-down modulation) how neuronal inputs are processed at the spinal trigeminal level [2]. Indeed, neuroimaging studies in humans have shown that, apart from early activation of the hypothalamus before a migraine attack: (i) an altered functional coupling with the spinal trigeminal nuclei during the attack phase occurs [16,20,21]; and (ii) a rostral brainstem and cortical (cingulate, prefrontal, insular and cerebellar cortices) activation also is observed [22]. Together, these data suggest that, although peripheral neurovascular events are relevant, supraspinal mechanisms (as triggers?) may play a key role in migraine pathophysiology [2]. In any case, current data suggest that the gepants and mAbs seem to act preferentially at the peripheral level [23,24].

2. An overview of current antimigraine pharmacotherapy

2.1. General considerations

The experimental strategies that have attempted to identify the mechanisms involved in the pathophysiology of migraine and their relevance in the progress of antimigraine pharmacotherapies have been extensively reviewed and updated over the years; to cite only a few selected references, see for

Table 1. A general migraine classification and associated symptoms according to the 3rd edition of the International Classification of Headache Disorders (ICHD-3). For further details, see [1].

MIGRAINE WITH AURA*	CHRONIC MIGRAINE
A. At least two attacks fulfilling criteria B and C B. One or more of the following fully reversible aura symptoms 1. <i>Visual</i> 2. <i>Sensory</i> 3. <i>Speech, language</i> 4. <i>Motor</i> 5. <i>Brainstem</i> 6. <i>Retinal</i> C. At least three of the following six characteristics 1. <i>One or more aura symptom spreads gradually (~5 min)</i> 2. <i>Two or more aura symptoms occur in succession</i> 3. <i>Each aura symptoms last for 5–60 min</i> 4. <i>At least one aura symptom is unilateral</i> 5. <i>At least one aura symptom is unilateral is positive</i> 6. <i>During or after aura (within 60 min) the headache occur</i> D. No better accounted for another ICHD-3 diagnosis	A. <i>Headache**</i> on ≥ 15 days per month for >3 months and fulfilling criteria B and C B. Occurring in a patient who has at least five attacks fulfilling criteria B-D for Migraine without aura and B and C for Migraine with aura C. On eight days/month >3 months fulfilling any of the following criteria: 1. <i>C and D for Migraine without aura</i> 2. <i>B and C for Migraine with aura</i> D. No better accounted for another ICHD-3 diagnosis
MIGRAINE WITHOUT AURA A. At least five attacks fulfilling criteria B-D B. Headache attacks lasting 4–72 h (untreated or unsuccessfully treated) (< 18 h in children and adolescents) C. Headaches have at least two of the following four characteristics: 1. <i>Unilateral location</i> 2. <i>Pulsating quality</i> 3. <i>Moderate or severe pain intensity</i> 4. <i>Aggravation by or causing avoidance of routine physical activity</i> D. During the headache, at least one of the following: 1. <i>Nausea and/or vomiting</i> 2. <i>Photophobia and phonophobia</i> E. No better accounted for another ICHD-3 diagnosis	General notes: * Migraine with aura can be subclassified into migraine: (i) with typical aura (with or without headache); (ii) with brainstem aura; (iii) hemiplegic migraine (familial hemiplegic migraine or sporadic hemiplegic migraine); and (iv) retinal migraine. ** Tension-type like or migraine-like headache

example [2–6,8,25–30]. Admittedly, it would be impossible to quote all migraine's historical details, therapeutic progress, as well as retrospective and prospective studies in this review. Nevertheless, suffice it to say that the last 80 years (and particularly the past four decades) have witnessed major advancements in both the understanding of migraine pathophysiology and the development of antimigraine pharmacotherapies. With these considerations in mind, only the most relevant pharmacological findings will be highlighted and summarized below.

2.2. Specific and nonspecific antimigraine pharmacotherapies

Since the trigeminovascular system plays a role in a migraine attack, any molecule (endogenous or exogenous) capable of interfering with the activity of this system could be used as an antimigraine drug [31]. Certainly, migraine is a type of pain and, consequently, it can be aborted with nonspecific analgesic medications (e.g., non-steroidal anti-inflammatory drugs; NSAIDs), at least when the attack ranges between mild-to-moderate pain [32]. Furthermore, in patients with recurrent migraine attacks, the use of nonspecific prophylactic agents is encouraged [33]; in this case, the use of β -blockers, Ca^{2+} channel blockers, and antiepileptic drugs are considered effective since they can reduce in ~50% the frequency of migraine attacks. Nevertheless, as previously reviewed [10], the

inadequate use of these nonspecific treatments can lead to several adverse events, ranging from gastrointestinal side effects to the induction of MOH in the case of NSAIDs, or autonomic (hypotension) and central (e.g., fatigue, insomnia, dizziness, depression, etc.) effects in the case of propranolol, flunarizine, and valproic acid. These undesirable effects result in poor compliance and adherence to the treatments [34].

The specific treatment of migraine started with the use of ergots (i.e., ergotamine and dihydroergotamine) and, subsequently, with the further development of more selective drugs (i.e., triptans) with an action upon the $5\text{-HT}_{1\text{B}/1\text{D}/1\text{F}}$ receptors subtypes. Currently, molecules targeting the $5\text{-HT}_{1\text{F}}$ receptors (i.e., lasmiditan) or the CGRPergic system (i.e., gepants and mAbs) have been approved. In general, the current specific pharmacotherapy targets the central and/or peripheral $5\text{-HT}_{1\text{B}/1\text{D}/1\text{F}}$ receptors and the CGRPergic transmission in the trigeminovascular system, and it can be broadly classified into: (i) acute abortive drugs, which can abolish an individual migraine attack (e.g., ergots, triptans, ditans and some gepants); and (ii) prophylactic drugs, which can prevent migraine attacks (e.g., mAbs against CGRP or its receptor, as well as the potential use of atogepant currently under U.S. Food and Drug Administration revision) (see Table 2 and Table 3 for details).

At this point, it is interesting to note that gepants and mAbs target the same system, but the current clinical classification as abortive or prophylactic molecules is based on the FDA-approval status. Certainly, as discussed below (see

Table 2. Ergots and triptans as specific acute antimigraine drugs, as well as their common and serious adverse events. *.

Ergots (nonselective $\alpha_{1/2}$ -adrenoceptors, D _{1/2} -like receptors, and 5-HT ₁ receptor ligands) **				
Drug	Affinity (pK_i or pK_D)	LogP	Routes of administration	Adverse events (>5%)
Ergotamine	5-HT _{1B} (pK_D : 8.7) 5-HT _{1D} (pK_i : 9.1) $\alpha_{1/2}$ -adr (pK_i : >8.0) D ₂ -like (pK_i : 8.5)	2.6–2.9	Oral, rectal	Nausea, vomiting, paresthesia and ergotism. To prevent MOH, their use must be limited to 10 days.
Dihydroergotamine	5-HT _{1B} (pK_D : 9.2) 5-HT _{1D} (pK_i : 8.6) $\alpha_{1/2}$ -adr (pK_i : 8.0) D ₂ -like (pK_i : 7.0)	3.0–2.7	i.v., nasal, oral, i.m., s.c.	
Triptans (5-HT _{1B/1D/1F} receptor agonists)				
Drug	Affinity (pK_i or pIC_{50} values for 5HT _{1B} /5HT _{1D} /5HT _{1F} receptors)	LogP	Routes of administration	Adverse events (>5%)
Sumatriptan	pK_i : 7.8/8.5/7.9	0.9	Oral, s.c., nasal	Chest discomfort, nausea, dizziness, distal paresthesia, fatigue. To prevent MOH, their use must be limited to 10 days. Between 15% and 40% of patients with oral triptans experienced a worsening of headache after pain free.
Zolmitriptan	pK_i : 8.3/9.2/7.2	1.8	Oral	
Naratriptan	pK_i : >8.1/>8.4/8.2	1.6	Oral	
Rizatriptan	pK_i : >6.9/>7.9/>6.8	1.4	Oral	
Almotriptan	pIC_{50} : 8.0/8.0/n.d.	1.5–2.0	Oral	
Eletriptan	pK_i : 8.0/8.9/8.0	3.9	Oral	
Frovatriptan	pK_i : 8.6/8.4/7.0	1.1–1.2	Oral	

* Part of these data were taken from [8,11,35–37].

** The pharmacology of 5-HT receptors is complex. Currently, these 5-HT family receptors are composed of seven types (5-HT₁ – 5-HT₇ receptors) and at least sixteen 5-HT receptor subtypes. With the exception of the ionotropic 5-HT₃ receptors, all are G protein-coupled receptors (GPCRs). In this sense, ergots are ‘promiscuous’ (*i.e.*, nonselective) drugs that display affinity (pK_i > 6) for the 5-HT₁ (agonist), 5-HT₂ (partial agonist), 5-HT₅ (antagonist), 5-HT₆ (agonist), and 5-HT₇ (antagonist) receptors. In the context of migraine and 5-HT receptors, the therapeutic action of ergots has been primarily related to activation of 5-HT₁ receptors (particularly, the 5-HT_{1B} and 5-HT_{1D} receptor subtypes). See [38,39] for an extensive revision on this subject.

Table 3. Some characteristics and properties of the recent FDA-approved antimigraine drugs. Lasmiditan and gepants are intended for acute treatment, whereas mAbs has been approved for prophylactic treatment. s.c., subcutaneous; i.v., intravenous.

Ditans (5-HT _{1F} receptor agonists)					
Drug	Affinity (pK_i)	Dose/route	$T_{max}/t_{1/2}$	logP	Adverse events (>2%)
Lasmiditan (Reyvow®)	8.7	50, 100 or 200 mg/oral	1.8/5.7	2.8–3.3	Dizziness, fatigue, paresthesia, sedation, nausea, vomiting, muscle weakness
CGRP receptor (CLR/RAMP1) antagonists					
Drug	Affinity (pK_i)	Dose/route	$T_{max}/t_{1/2}$	logP	Adverse events
Ubrogepant (Ubrovelvy®)	10.2	50 or 100 mg/oral	2 h/5–7 h	3.1	Nausea, somnolence and dry mouth (>2%)
Rimegepant (Nurtec®)	10.6	76 mg/oral	1.5 h/~11 h	2.7–2.9	Nausea (>1%)
Monoclonal antibodies against CGRPergic transmission*					
Drug	Affinity (pK_D)	Doses/route	$T_{max}/t_{1/2}$	IgG Type	Adverse events (>2%)
Galcanezumab (Emgality®)	8.7 (against α -CGRP)	120 mg (m)/s.c. 300 mg (m)/s.c.	7–14 d/27 d	Humanized IgG ₄ , mAb from CHO	Injection site pain (including erythema), an increase of upper respiratory tract infections and abdominal pain
Fremanezumab (Ajovy®)	>10.3 (against α -CGRP)	225 mg (m)/s.c. 675 mg (q)/s.c.	3–20 d/32 d	Humanized IgG ₂ , mAb from CHO	Injection site pain and sinusitis
Eptinezumab** (Vyepti®)	10.0 (pEC_{50}) (against α -CGRP)	100 mg (q)/i.v. 300 mg (q)/i.v.	4.8 h/26 d	Humanized IgG ₁ , mAb from <i>Pichia pastoris</i>	An increase of upper respiratory tract infection, urinary tract infection, fatigue, back pain, and arthralgia
Erenumab*** (Aimovig®)	10.7 (pK_i) (against CGRP receptor)	70 mg (m)/i.v. 140 mg (m)/i.v.	3–14 d/28 d	Human IgG ₂ , mAb from CHO	Nasopharyngitis, fatigue, and headache. At least 3% of patients developed neutralizing antibodies

Part of these data were taken from [8,11,187–190].

* Erenumab is considered a fully human immunoglobulin whereas galcanezumab, fremanezumab and eptinezumab are considered as humanized immunoglobulins (see [30] for references).

** Note that in contrast to other mAbs, eptinezumab reaches its T_{max} in ~ 5 h, whereas the others need more than 3 days. The reason for this difference is the route of administration.

*** >5000-fold more selective for CGRP receptor than other human calcitonin receptor (*i.e.*, AM₁, AM₂, AMY₁ and CT receptor).

Section 4), the pharmacokinetics of these compounds play a role in the clinical setting; *i.e.*, the mAbs are less frequently (~monthly) administered than the gepants (~hours) due to the fact that mAbs have much longer half-

lives (see Table 3 for details). Hence, small molecules (*i.e.*, the gepants), but not protein drugs (*i.e.*, the mAbs), may induce migraine overuse headache if used for a long time. In addition, considering the hepato-toxicity effects observed

with the 1st generation of gepants (*i.e.*, telcagepant, MK-3207 and probably BI 44370TA) [11,30], the chronic use of current FDA-approved 2nd generation of gepants (ubrogepant and rimegepant) is discouraged. We must acknowledge that the chemical structures of the 1st and 2nd generations of gepants are quite different lacking the potential hepatotoxicity effects. Indeed, a current Phase III randomized clinical trial (RCT) is evaluating the prophylactic effect of atogepant [40]; moreover, intranasal formulations of a new highly liposoluble gepant, *i.e.* zavegepant (formerly known as vazegepant or BHV-3500) are in the pipeline (Phase II/III RCT) [41].

In general, all migraine patients require acute treatment, whereas only patients that suffer from frequent migraine attacks require, in addition to acute treatment, prophylactic treatment by drugs taken daily to reduce the number and/or severity of migraine attacks [33].

2.3. The site of action of current specific antimigraine drugs

Although migraine is currently considered a neurovascular disorder, it was first devised as a vascular disease (Thomas Willis, 1664) where vasodilatation within the head induces pain. Then, in 1873, Liveing suggested a neuronal theory where 'optic' thalamus plays a role (for a brief historical perspective see [42]). The vascular theory was later (1940's) supported by the seminal observations of Harold Wolff [43,44], describing that migraine pain can be relieved by physical compression of the temporal branches of the external carotid

artery or by ergotamine or dihydroergotamine [45]. Certainly, ergots act on an array of receptors ($\alpha_{1/2}$ -adrenoceptors, serotonin 5-HT_{1/2/5/6/7} receptors, and dopamine D_{1/2}-like receptors) and, consequently, several side effects, ranging from nausea, vomiting, and cardiovascular events, are induced [46]. Further research on the mechanisms involved in the ergots' effects led to the establishment of the role of serotonin (5-HT) and craniovascular 5-HT_{1B/1D} receptors in the pathophysiology of migraine. These findings fostered the development of selective 5-HT_{1B/1D} receptor agonists with preferential vasoconstrictor effects on cranial blood vessels; these compounds (*i.e.*, triptans) were successfully incorporated in migraine treatment in the 1990's.

At this point, we must keep in mind that before and after a migraine attack, some neurological symptoms (*e.g.*, mood changes, nausea, photophobia) are observed, pointing out the role of neural mechanisms as described by Lance *et al.* [47] and Moskowitz [48]. In this context, although at early stages the vascular effects of ergots and triptans were considered relevant for its therapeutic effects [49], these compounds also modify the neuronal activity at peripheral and central levels (Figure 1). Briefly, and apart from their vascular effects, ergots and triptans act at neuronal level by [50–54]:

- (i) **Peripheral** inhibition of CGRP release from trigeminal axonal nerve endings, which results in a diminution of the vasodilatation induced by this neuropeptide. This action seems to be related with prejunctional activation of 5-HT_{1B/1D/1F} receptors and $\alpha_{2A/2C}$ -adrenoceptors [50,52,53].

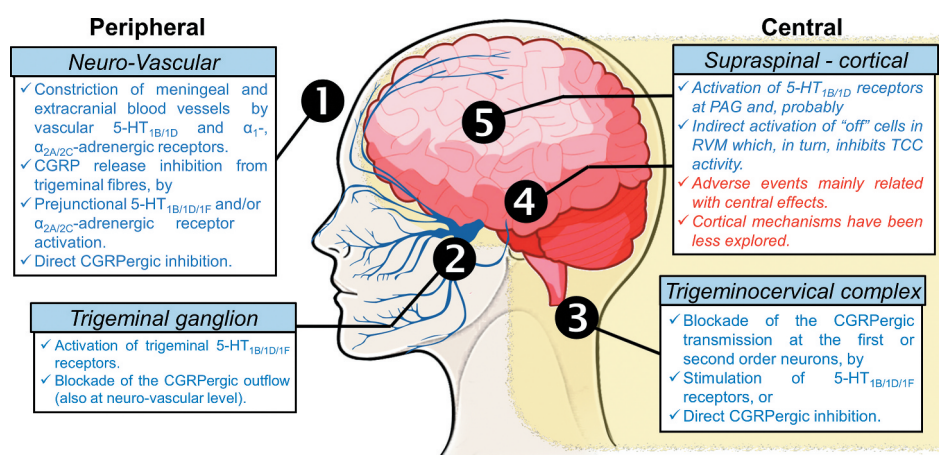


Figure 1. The site of action of specific antimigraine drugs, peripheral vs. central mechanisms. Although before, during, and after a migraine attack several neuromediators are involved, it has been suggested that serotonin and CGRP play a preponderant role. It is well-established that during the headache phase, a decrease of serotonergic transmission and an increase in CGRP release from the perivascular trigeminal nerve is present. In conjunction, these phenomena lead to vasodilatation, which in turn produces sensitization of the trigeminal afferents resulting in overstimulation of the TCC. Additionally, at this level, an increase of the CGRPergic transmission seems to be present. From this data, agonists at 5-HT_{1B/1D/1F} receptors, blockers of CGRP receptors or CGRP scavengers are the current specific antimigraine drugs. Ergots and triptans not only induce a direct vasoconstriction (mediated by $\alpha_{2A/2C}$ -adrenergic and/or 5-HT_{1B/1D} receptors stimulation), but they also inhibit CGRP release by prejunctional mechanisms at dura mater, trigeminal ganglion, and trigeminal nucleus caudalis. In the case of lasmiditan (a selective 5-HT_{1F} receptor agonist), this drug is devoid of any effects on vascular smooth muscle, and its therapeutic effect is mainly related to inhibition of CGRP release at the peripheral and central levels. In contrast, CGRP receptor antagonists and mAbs act directly by inhibiting the CGRPergic transmission; in the case of mAbs, their therapeutic effect seems to be related with an action at peripheral, rather than central, levels. At supraspinal levels, triptans may act by enhancing descending antinociceptive mechanisms by recruitment of the PAG-RVM-TCC structures. Furthermore, although cortical spreading depression (CSD) seems to play a role in migraine with aura, the effect of specific antimigraine drugs at the cortical level has been less explored; nevertheless, some data suggest that MOH induced by triptans can disrupt the brain network homeostasis facilitating CSD. Abbreviations: CGRP, calcitonin gene-related peptide; CSD, cortical spreading depression; MOH, migraine overuse headache; PAG, periaqueductal gray matter; RVM, rostral ventromedial medulla; TCC, trigeminocervical complex.

- (ii) **Peripheral** actions diminishing the probability of membrane depolarization of trigeminal ganglia neurons and, consequently, an inhibition of peripheral and central sensitization. Activation of 5-HT_{1B/1D/1F} receptors is involved [51,54].
- (iii) **Central** inhibition of the peripheral nociceptive-vascular transmission (reflex?) at the trigeminocervical complex (TCC) level. In this case, by activation of 5-HT_{1B/1D/1F} receptors and $\alpha_{2A/2C}$ -adrenoceptors probably located at the 1st or 2nd order neurons [52,53].
- (iv) **Central** enhancement of the activity of descending analgesic systems. For example, microinjection of naratriptan (a 2nd generation triptan) in the periaqueductal gray matter (PAG) inhibits dural nociceptive input at TCC [55], suggesting a descending modulation of trigemino-vascular nociceptive transmission probably by activation of 'off' cells at the rostral ventromedial medulla (RVM).

Besides, the most common adverse event in the case of ergots is related with its nonspecific vascular actions [56]. In this respect, the use of triptans certainly represents a therapeutic breakthrough, but their vasoconstrictor side-effects on coronary blood vessels could limit their use in patients with coronary artery disease [57]. In this respect, MaassenVanDenBrink et al. [57] showed in *in vitro* experiments that the ergots and triptans used as antimigraine agents are capable of inducing vasoconstriction of the human coronary artery; nevertheless, in subjects with normal coronary circulation the plasma concentrations of these compounds after oral intake probably do not reach levels to induce myocardial ischemia.

Furthermore, as previously reviewed [11], the triptans (particularly the 2nd generation triptans) not only induce acute central side-effects (e.g., dizziness, nausea, fatigue), but their recurrent use may promote migraine chronification which, in turn, could lead to MOH, a secondary headache disorder associated with an atypical structure and function in several regions of the central nervous system (CNS) related with pain processing and addiction [58,59].

In the case of lasmiditan (a selective 5-HT_{1F} receptor agonist), its therapeutic effect has been related with neuronal actions at the perivascular nerve fibers, trigeminal ganglia and at TCC level [60]. Certainly, lasmiditan was developed to avoid the vascular side effects produced by the triptans, in view that 5-HT_{1F} receptors are poorly expressed in vascular smooth muscle. Indeed, in comparison with sumatriptan, lasmiditan has no vasoconstrictor effects on the middle meningeal artery or human coronary arteries [61]. Nevertheless, the probability to observe undesirable central side effects (e.g., dizziness, nausea, fatigue) by this compound seems to be high in comparison with the triptans [62,63]. Most importantly, a recent report [64] suggests that lasmiditan has the potential to develop MOH. Together, these data may imply that central 5-HT_{1B/1D/1F} receptors, which are expressed in several areas of the brain [65], play a role in MOH development.

Finally, the most recent groundbreaking therapeutic development against migraine is represented by the mAbs (*i.e.*,

galcanezumab, fremanezumab, eptinezumab and erenumab) and a pair of gepants (*i.e.*, rimegepant and atogepant), which have recently been approved for antimigraine treatment [4,8,11]. In general, these molecules can interfere with the CGRPergic transmission and seem to be effective, well tolerated and with fewer side effects, probably because their actions are related with peripheral, rather than central, effects [4,8,11]. Interestingly, and contrasting with the triptans and ditans, blockade of the CGRPergic transmission does not result in MOH [65,66]. Indeed, the mAbs have a very high molecular weight (~150 kDa) and cannot cross the blood-brain barrier (BBB), implying a peripheral site of action; whereas in the case of the gepants (molecular weight < 0.9 kDa), further studies on their site of action are required [4,8,11]. However, a previous report using telcagepant showed that after *i.v.* administration no relevant receptor occupancy to exert a pharmacological effect is achieved in the CNS [23]. The lack of central side effects of these new treatments tends to support the idea that their antimigraine actions are confined to the periphery.

The above findings lead us to pose the question: if an array of molecules to treat migraine exists, why do we need new perspectives for the development of novel pharmacotherapies against this disorder? The shortest answer may hinge on the conundrum that not all patients respond to the current antimigraine treatments. For example, although triptans are the gold standard in migraine treatment, only ~20% of patients with episodic migraine use triptans (at least in the U.S.A [67]). Furthermore, some other issues may impact on the overall adherence to treatment, namely: (i) the need for an early intake (during the aura phase) to abort a migraine attack; (ii) headache recurrence; (iii) >30% patients failed to respond to treatment; (iv) a probability to develop MOH due to misuse; and (v) half of the patients present nausea and other adverse events [11,68–71]. From these data, it is evident that new targets (see below) and/or new pharmaceutical developments (e.g., intranasal dihydroergotamine [45]) are necessary to improve antimigraine treatment.

3. CGRP, PACAP, oxytocin and orexins in migraine pathophysiology

3.1. An overview

Since the appearance of CGRP in migraine pathophysiology [72], a renewed interest in the role of other neuropeptides has emerged. Indeed, in the last decade, and apart from CGRP, other neuropeptides from hypothalamic tissue [*i.e.*, pituitary adenylate cyclase-activating peptide (PACAP), oxytocin (OT) and orexins] with a potential role in the pathophysiology of migraine has arisen. Certainly, one of the most interesting features of these molecules is their capability to exert, overall, pleiotropic effects in the CNS and the periphery; hence, they may act as neuromodulators by enhancing or dampening the synaptic activity evoked by classical neurotransmitters [73,74]. Furthermore, due to their chemical nature, not only can they exert their effects in the

synapse (where they are released from dense core vesicles), but they can also travel relatively long distances (mm) to exert paracrine effects [73].

3.2. Calcitonin gene-related peptide (CGRP)

In the case of CGRP, it is undeniable that this peptide plays a key role in migraine pathophysiology [5,72]. Briefly, CGRP is a 37 amino acid neuropeptide with potent vasodilator properties [75] that: (i) seems to play a role in vascular modulation [76]; (ii) is related to the calcitonin gene peptide super family; and (iii) consists of two isoforms (*i.e.*, α -CGRP and β -CGRP) [76–78]. α -CGRP is predominantly located in peripheral and central sensory neurons [78], whereas β -CGRP can be mainly found in the pituitary gland and the enteric nervous system [76,77]. The effect of both isoforms can be antagonized by CGRP receptor antagonists (gepants) or blocked by mAbs against CGRP (see below).

Among other important characteristics, CGRP: (i) is primarily located in C-fibers, and less predominantly in A δ -fibers, which have an extensive perivascular presence [79,80]; (ii) is extensively distributed in the central and peripheral nervous systems [11,81]; and (iii) produces its effects by interacting with at least one receptor (*i.e.*, the CGRP receptor), belonging to the 7-transmembrane domain receptors (the G-protein coupled receptors, GPCRs) [82].

Several lines of evidence have suggested that, under physiological conditions, CGRP does not appear to participate significantly in the modulation of vascular tone [76,78,83], but it has been implicated in the pathophysiology of migraine and other vascular disorders, including preeclampsia and hypertension [5,81,84]. Indeed, the release of CGRP from activated trigeminal sensory nerves produces, via activation of CGRP receptors, intra/extra cranial vasodilatation and modulation of vascular nociception at the central level [5,72]. These findings, which involve CGRP in migraine pathogenesis, have led researchers to propose that the interference with the CGRPergic transmission may result in antimigraine action [5,11,81,84].

CGRP and its receptor are found in the trigeminovascular system, as well as in other supraspinal structures relevant to migraine pathophysiology [5]. In contrast to many GPCRs, the CGRP receptor is a heterodimer of 2 proteins, namely: (i) the receptor activity-modifying protein 1 (RAMP1), a single-transmembrane-domain protein; and (ii) the calcitonin receptor-like receptor (CLR), the GPCR protein [78,85] (see Figure 2). An additional third protein, the receptor component protein (RCP), is required to form an optimally functional CGRP receptor, which is coupled to the intracellular heterotrimeric G_s proteins [86]. At this point, we must keep in mind that CGRP can activate the adrenomedullin (AM) receptors (particularly the AMY₁ receptor) in view of the similarities between the CGRP and AM receptors (*i.e.*, both receptors require the RAMP1 protein) [87].

Within the context of migraine pathophysiology, it is noteworthy that: (i) experimental activation of the trigeminal ganglia in animals and humans results in CGRP release [88]; (ii)

plasma and cerebrospinal fluid levels of CGRP are increased during migraine [19,89,90]; (iii) injection of CGRP induced a delayed, migraine-like headache in migraineurs with and without aura [91,92]; and (iv) sumatriptan (a 5-HT_{1B/1D/1F} receptor agonist), the gold standard in migraine treatment, normalized the increased plasmatic CGRP levels [93]. The fact that CGRP failed to induce migraine in healthy controls suggests that migraineurs have an increased sensitivity to CGRP [30]. From this perspective, it comes as no surprise that the central (mainly at TCC) and peripheral actions of CGRP are well positioned to contribute to migraine symptoms.

3.3. Hypothalamic neuropeptides and migraine pathophysiology

The hypothalamus has been thought to play an important role in the genesis of some of the most common symptoms that appear before, during and after the onset of the headache phase of migraine [47]. In general terms, the hypothalamus: (i) is connected with several structures of the CNS and also with the vasculature due to the hypothalamus-pituitary axis; and (ii) serves as a brain-hormonal interface through which the brain controls peripheral body systems [94]. Besides the regulation of body temperature, sleep, food and water intake, it exerts a specific control over the autonomic nervous system and cyclic phenomena such as circadian and circannual body rhythms [94].

On this basis, and considering that the hypothalamus contains several neuropeptides with antinociceptive/analgesic action, Alstadhaug [95] points out the potential role of this structure in the pathophysiology of migraine pain. Indeed, the studies of Schulte *et al.* [20] and Denuelle *et al.* [96] using functional (magnetic resonance imaging) MRI or positron emission tomography (PET) strongly support the role of the hypothalamus during a migraine attack. Accordingly, some of the symptoms like nausea, photo- and phonophobia have been related with a dysfunction of sensory processing not only at the brainstem, but also at the hypothalamus level [2,16,96–98].

Certainly, the accompanying symptoms of nausea, photo- and phonophobia may sometimes be simultaneous to the headache, but they usually start before the headache phase [2,5,16]. In addition, most migraine patients experience at least in some of these attacks, migraine premonitory symptoms such as changes in appetite (food craving or nausea), and sleep-walking rhythms (yawning, fatigue, sleep disturbances), hypersensitivity to certain stimuli (photo-, osmo-, phonophobia), mood changes, and changes in liquid tolerance and others [99–102].

Furthermore, during the headache phase, migraine appears to be an isolated event of headache, but an oscillation of sensory and bodily functions exists where the headache appears to be just one of many symptoms [103–105]. Finally, patients experience a postdromal phase (occurring after headache remission), characterized by fatigue, tiredness, euphoria or dysphoria, but also alterations in appetite or liquid tolerance [106–108]. These findings not only depict the symptoms

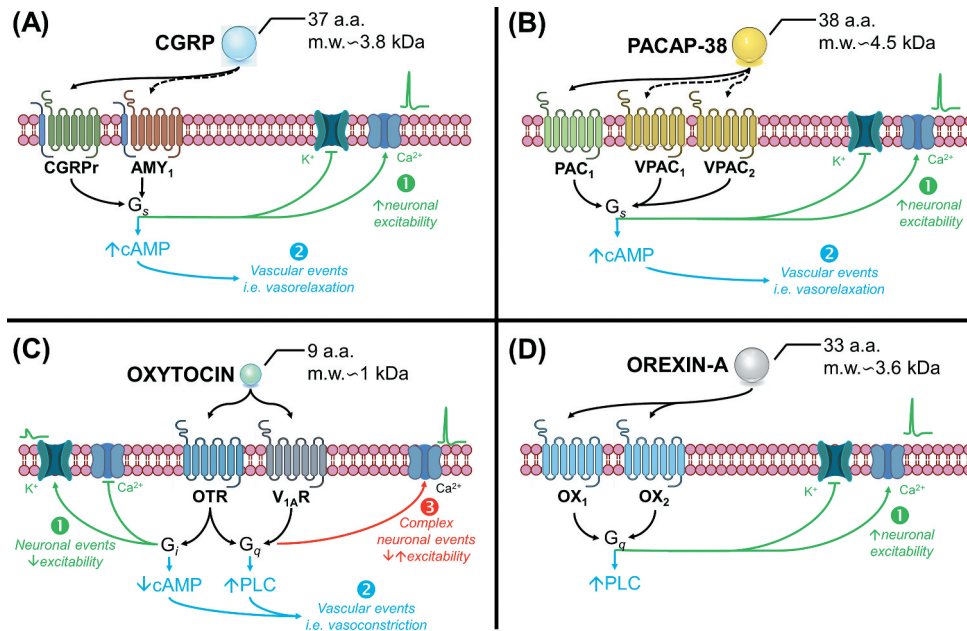


Figure 2. Main intracellular events elicited by CGRP, PACAP-38, OT, and orexin-A upon interaction with their respective GPCRs. (A) CGRP has affinity for the CGRP receptor and the AMY_1 receptors; upon binding, the G_s protein activates AC, which catalyses the conversion of ATP to cAMP which, in turn, activates PKA to regulate the activity of numerous intracellular processes, including K^+ channels, L-type Ca^{2+} channels, and CREB protein. Both receptors are expressed in blood vessels, as well as at the level of trigeminal neurons (at ganglia and nucleus); the downstream effects of activation of these receptors include neuronal excitability, neurotransmitter release, NO production, and vasorelaxation. Note that the CGRP receptor and the AMY_1 receptors are considered as atypical GPCRs in view that, apart from the 7-transmembrane protein (the CLR), it requires an accessory protein to be functional, namely RAMP1. (B) PACAP-38 can bind to PAC_1 , $VPAC_1$, and $VPAC_2$ receptors. Upon activation, these receptors cause similar intracellular signaling and effects to those elicited by CGRP receptors. PAC_1 receptors are present in the trigeminovascular system, hypothalamus, sphenopalatine ganglia, as well as in cerebral and meningeal arteries. (C) OT binds to the OT receptor and $V_{1A}R$; both receptors are expressed in central and peripheral tissues (including blood vessels). The OT receptor can functionally associate with G_q proteins, but some studies suggest that a functional coupling with G_i proteins may also occur. The OT receptor coupled to G_q stimulates Ca^{2+} signaling and increases PLC leading to neuronal or vascular (vasoconstriction) events. Moreover, OT receptor coupled to G_i decreases cAMP, facilitates K^+ efflux, and closes Ca^{2+} channels. In general terms, it has been suggested that activation of OT receptor induces inhibition of nociceptive transmission, but the exact nature of G_q or G_i proteins remains unsolved (at least in *in vivo* experiments). (D) Orexin-A binds to OX_1 or OX_2 receptors; activation of both receptors results in elevated intracellular Ca^{2+} levels mediated by G_q and PLC activation. The disruption of these neuropeptidergic systems can contribute to altered homeostatic mechanisms that can influence a migraine attack susceptibility, premonitory and associated symptoms, and migraine nociception. These systems have been identified as a potential therapeutic target for migraine. Abbreviations: AC, adenylate cyclase; AMY_1 , adrenomedullin type 1 receptor; CLR, calcitonin-like receptor; CREB, cAMP response element-binding; CT, calcitonin receptor; GPCR, G protein-coupled receptor; m.w., molecular weight; NO, nitric oxide; OT receptor, oxytocin receptor; PAC_1 , PAC_1 receptor; PACAP, pituitary adenylate cyclase-activating peptide; PKA, protein kinase A; PLC, phospholipase C; RAMP1, receptor activity modifying protein 1; $V_{1A}R$, the vasopressin 1A receptor; $VPAC_{1/2}$, VIP and PACAP receptors; VIP, vasoactive intestinal peptide.

as a continuum during a migraine attack but, coupled to the neuroimaging studies [109–111], the role of the hypothalamus in migraine pathophysiology is strongly supported.

From an anatomical perspective, the hypothalamus also has various neuroanatomical connections to pain modulating systems including the spinal trigeminal nuclei [112]. The hypothalamus receives sensory inputs from areas innervated by the trigeminal nerve via the trigemino-hypothalamic tract and, in return, it can modulate the neuronal activity of the trigeminal nucleus caudalis via the posterior part of the hypothalamus [113]. In this context, in the last decade, the role of hypothalamic neuropeptides such as PACAP, OT and orexins has emerged as an interesting concept in which other neuropeptides (apart from CGRP) can modulate the trigeminovascular system.

3.3.1. Pituitary adenylate cyclase-activating peptide (PACAP)

PACAP is a major sensory and vasodilator neuropeptide involved in parasympathetic neurotransmission to the cranial vasculature [114,115]. This neuropeptide, which belongs to

the glucagon/secretin superfamily of peptides together with vasoactive intestinal peptide (VIP) [116], exists in two bioactive isoforms, namely: a 38-amino acid form (PACAP-38) and a truncated 27-amino acid form (PACAP-27) [117]. PACAP-38 is present in first-order neurons of the trigeminal ganglion [118], in second-order neurons in the TCC [119], and, in the parasympathetic sphenopalatine ganglion [120]. Moreover, PACAP-38 is the most predominant isoform expressed in the brain, and it has also been identified in the cerebral cortex, cerebellum, brainstem, and hypothalamus [121], as well as in the cerebral and meningeal arteries [122].

In general, the effect of PACAP-38 is mediated through three GPCRs, namely, PAC_1 , $VPAC_1$, and $VPAC_2$ [123] (see Figure 2). Furthermore, it is pharmacologically interesting to note that PACAP-38 and VIP display a similar affinity for $VPAC_1$ (pK_i : 8.2 and 8.5–9.8, respectively) and $VPAC_2$ (pEC_{50} : 7.3–9.3 for both peptides) receptors, whereas in the case of PAC_1 receptor, PACAP-38 has a much higher affinity than VIP (pK_i : 8.4 vs. 6.0) [124]. These findings could have several repercussions on the specificity of new molecules at the former receptors, and the implications of VIP in migraine. Upon activation, these receptors

cause downstream production of cyclic adenosine monophosphate (cAMP) through adenylate cyclase (AC) stimulation [125].

Regarding migraine, upon stimulation of the superior sagittal sinus (SSS) or trigeminal ganglion, an increase in PACAP release is induced [126,127], whereas intravenous infusion of PACAP-38 induces cranial vasodilatation and also a delayed-type migraine-like attack [128]. Considering that during a migraine headache, the increase of PACAP in the external jugular vein can be reduced with sumatriptan, a role for this neuropeptide in migraine pathophysiology is suggested [129]. Indeed, closely related to this suggestion, Amin *et al.* [130] analyzed the effect (in patients with migraine without aura) of PACAP-38 in extra- and intracranial arteries using magnetic resonance angiography; they found that PACAP-38-induced migraine was associated with dilatation of extracranial arteries. Moreover, this neuropeptide was found: (i) to disrupt the brain network connectivity [131]; and (ii) to trigger CGRP release in the trigeminal nucleus caudalis [132], which further supports a cooperative action of these neuropeptides in migraine pathophysiology. These lines of evidence have raised interest in the search for specific treatment options targeting the PACAP-38 molecule or its PAC₁ receptor [133].

Interestingly, at this point, Robert *et al.* [134] showed that microinjection of PACAP-38 in the hypothalamic paraventricular nucleus (PVN) enhanced the spontaneous neuronal activity of TCC neurons; this facilitation is blocked by pre-treatment with PACAP-(6-38) (a PAC₁ receptor antagonist) or naratriptan, both given at the PVN level. Indeed, a neuronal projection between the PVN and the superior salivatory nucleus (a region that gives rise to parasympathetic outflow to cephalic and ocular/nasal structures) was described [134]. These data suggest that the PVN modulates directly the outflow at TCC [113], and also that it directly regulates the parasympathetic outflow [134].

More recently, Akerman and Goadsby [135] suggested that central, but not peripheral, PAC₁ receptors mediate the sensitization of trigeminocervical neurons. Hence, a role of high order processing pain has been suggested, and this suggestion is reinforced when considering that: (i) PAC1 receptor-deficient mice exhibited an increase of physical withdrawal and exacerbated central sensitization during visceral pain [136]; and (ii) at the central nucleus of the amygdala, PACAP-38 decreased latency and threshold responses to thermal and mechanical stimuli [137].

3.3.2. Oxytocin (OT)

Oxytocin (OT) is a neurohypophyseal nonapeptide, which plays an important role in the central and peripheral nervous systems [138]. Although best known for its function in lactation, uterine contraction during parturition and, more recently, for its role in social behaviors [139], a renewed interest in it as an analgesic molecule has emerged [140,141]. At this point, we must acknowledge that this peptide is involved in a plethora of effects in the peripheral and CNS [138], including anxiety, sociability, and drug abuse, among other functions [142,143].

Regarding its antinociceptive effects, this endogenous molecule exerts potential analgesic effects by modulating

nociception at peripheral, spinal, and supraspinal sites in the nervous system during physiological and pathological pain conditions [144–147]. Canonically, oxytocin activates its cognate receptor (the OT receptor), a seven-transmembrane receptor preferentially coupled to G_q proteins which, in turn, mobilizes intracellular Ca²⁺ leading to an array of metabotropic effects [148]; notwithstanding, coupling to G_i proteins has also been described and, consequently, biased signaling may play a key role in drug actions targeting this system [149]. Indeed, although much effort has been made to elucidate the mechanisms/pathways that OT engage to induce its effects, the key downstream mechanisms remain unsolved (for refs. see [141,150,151]). *Grosso modo*, we know that, at the spinal level, OT inhibits nociception by enhancement of: (i) GABAergic activity; (ii) inhibition of transient potassium current (I_A); (iii) desensitization of spinal TRPV1 channels [151]; and (iv) disruption of the NMDA-evoked coordinated neuronal network activity [152].

From a physiological perspective, OT modulates the nociceptive input by activating descending endogenous mechanisms [144,146]. Indeed, experimental pain models that resemble some pathophysiological aspects of chronic pain showed that intrathecal OT relieves this type of pain, pointing out the relevance of oxytocinergic mechanisms modulating the nociceptive stimuli during pathological conditions [153–155]. Interestingly, some of these experimental results have been replicated in humans [156–158].

In the context of migraine, some short reports in humans suggest that, when given by intravenous [159] or intranasal route [160,161], this neuropeptide exerts an antimigraine action. Moreover, since a direct projection from PVN to nucleus caudalis exists [113], a role for OT in the modulation of the trigeminal nucleus is suggested. This modulation seems to be mediated by oxytocinergic mechanisms in view that OT microinjection at TCC inhibits the trigeminovascular nociception by OT receptor activation [162,163]. Similar results have been recorded at the spinal L4-L5 level in electrophysiological recordings of wide dynamic range cells [164]. Accordingly, in the human spinal trigeminal nucleus, the presence of OT receptors, but not vasopressin V_{1A} receptors (V_{1A}R), has been demonstrated [165]. This point is relevant in view that OT has a similar affinity for OT receptors (pK_i: 8.2–9.6) and V_{1A}R (pK_i: 6.9–8.3), and that the latter has been linked to cardiovascular regulation; consequently, potential undesirable effects can be elicited.

Finally, the potential role of OT as an antimigraine molecule gains weight when considering that: (i) intra-ganglionic (trigeminal) OT administration inhibits the mechanical hypersensitivity induced by injury of the infraorbital nerve [166]; (ii) OT inhibits the capsaicin-induced CGRP release from perivascular meningeal fibers [167]; and (iii) OT receptors are expressed in the trigeminal ganglia, and colocalize with CGRPergic fibers [167].

3.3.3. Orexin-A and orexin-B

Orexin-A and orexin-B, also known as hypocretins, are synthesized in the perifornical area in the posterior lateral hypothalamus and send dense projections to the PVN and other brain

structures (e.g., locus coeruleus, the tuberomammillary nucleus, raphe nucleus and the laterodorsal and pedunculopontine nuclei) [168,169]. Both orexins are mainly involved in sleep/wakefulness regulation [169], but they have also been associated with the feeding regulation [170,171] and are considered as an autonomic nervous system regulator [169–171]. The effects of these orexins are mediated by two distinct GPCRs coupled to G_q proteins, namely OX_1 and OX_2 receptors [171]. Within the context of migraine, orexin-A and orexin-B levels are lower in the periphery during episodic migraine, whereas orexin-A levels are increased in chronic migraine [172]. Interestingly, the intravenous administration of orexin-A, but not orexin-B, can inhibit dural evoked trigeminovascular transmission [173], but when locally injected into the posterior hypothalamus, orexin-A and orexin-B differentially modulate dural and cutaneous evoked nociceptive trigeminovascular responses (i.e., orexin-A inhibits the evoked neuronal responses in the TCC, while orexin-B facilitates these responses) [174]. Furthermore, since the hypothalamic micro-injection of orexin-A inhibited the spontaneous and evoked TCC activity, and this response was blocked by PAG micro-injection of GR127935 (a $5-HT_{1B/1D/1F}$ receptor antagonist), a descending orexinergic mechanism mediated by $5-HT_{1B/1D}$ receptors is suggested [175]. Given the role that both the hypothalamus and the orexinergic system play in the modulation of feeding, arousal and sleep, these data provide a causal link between the possible origin of the migraine trigger and the premonitory symptoms (hypothalamus) [10,98,176].

4. Current and prospective pharmacotherapies interacting with the CGRP, PACAP, oxytocin and orexin systems

4.1. The case of CGRP

Given the key role of CGRP in migraine pathophysiology [5,72], the last two decades have witnessed the development of molecules interfering with the CGRPergic transmission. Among the mechanisms of action of these antimigraine agents, which directly block CGRP or its receptor, a reduction on neurogenic inflammation stands out [177]; this, in turn, would result in migraine pain relief and a significant reduction in the rate and/or intensity of migraine attacks [6]. These novel antimigraine agents targeting CGRP or its receptor have been classified into [8,9]: (i) acute (i.e., gepants, which are potent non-peptide CGRP receptor antagonists); and (ii) prophylactic antimigraine drugs (i.e., human(ized) mAbs which directly block CGRP or its receptor).

4.1.1. Gepants, the oral CGRP receptor antagonists

The historical aspects describing the rationale and development of the gepants will not be addressed in detail here, as they have been extensively reviewed in many publications (e.g., see [4,5,8,11,27,28,30,178]). Hence, this section will mainly deal with the most recent developments.

As recently reviewed by Lambru *et al.* [179] and de Matteis *et al.* [180], the so-called ‘gepants’ encompass a class of small

molecule CGRP receptor antagonists including (in chronological order of development) olcegepant, telcagepant, MK3207, BI44370, ubrogepant, rimegepant and atogepant. These drugs: (i) are relatively small molecules (as compared to mAbs) with a molecular weight < 0.9 kDa; and (ii) block the CGRP receptors ($pK_i > 9.0$) located in the trigeminal ganglion and blood vessels (including those relevant in migraine pathophysiology, i.e., cranial blood vessels) [4,8,11].

The further development of the first four gepants was halted for several reasons, including, among others, pharmacokinetic issues and hepatotoxicity. Fortunately, a second generation of gepants, including ubrogepant, rimegepant, and atogepant were: (i) further developed; (ii) shown to be effective and safer than other antimigraine drugs in clinical trials; and (iii) suggested to have their main target outside the BBB [4,8,11]. Moreover, it is worthy of note that: (i) ubrogepant [181] and rimegepant [182] are now FDA-approved for the acute treatment of migraine; (ii) rimegepant is presently being evaluated in a clinical trial for the prophylactic treatment of migraine [183]; and (iii) atogepant, which displays a higher potency and longer half-life than ubrogepant, was specifically developed for migraine prophylaxis [184].

4.1.2. Human(ized) monoclonal antibodies (mAbs)

The mAbs against CGRP or its receptor were developed to complement the antimigraine strategy behind the gepants [4,8,11,30]. Since their plasma half-life is much longer than that of the gepants (see Table 3), they were proposed for migraine prophylaxis [185]. Moreover, based on their high molecular weight (~150 kDa): (i) the relevance of their central actions can be excluded as they do not seem to cross the BBB; (ii) liver toxicity can be ruled out as they are not metabolized by hepatic pathways; and (iii) their actions in the trigeminovascular system are predominantly mediated by peripheral mechanisms [4,8,11,30].

As previously reviewed in detail [4,8,11,30], the mAbs that directly block CGRP or its receptor represent a novel therapeutic strategy for migraine treatment. Indeed, four mAbs are presently FDA-approved for the prophylactic treatment of migraine, namely, fremanezumab, galcanezumab, eptinezumab, and erenumab [4,8,11,30]. These mAbs, which directly block CGRP (i.e., the humanized mAbs fremanezumab, galcanezumab, and eptinezumab) or the CGRP receptor (i.e., the fully human mAb erenumab): (i) are effective and safe (in the short-term) for migraine prophylaxis; and (ii) appear to produce fewer or no side effects as compared to those produced by the already established acute and prophylactic antimigraine drugs [4,8,11,30].

4.1.3. An overview of pharmacokinetics

As far as metabolic pathways are concerned, the mAbs are hydrolyzed by renal proteolysis into small peptides and amino acids. Since protein metabolism does not generate toxic mediators, these compounds are not hepatotoxic [8,11,30]. The main pharmacodynamic improvement of mAbs is their high affinity ($pK_i > 9.0$) and selectivity against their target [8,11,30],

whereas their pharmacokinetic improvement is their long elimination half-life (see Table 3).

4.1.4. Side effects associated with CGRP (receptor) blockade

As previously reviewed by Rivera-Mancilla *et al.* [8] and González-Hernández *et al.* [11], a significant concern generated by the long-term use of CGRP blockers for migraine prophylaxis (in particular the mAbs) is that CGRP acts physiologically as a potent vasodilator. Hence, it plays a role in the homeostasis of the gastrointestinal [186] and cardiovascular [84] systems under pathological conditions. The specific side effects produced by every one of the mAbs have been summarized in other reviews [8,11,30]. Hence, we will only highlight here, overall, that the long-term CGRPergic blockade could cause a loss of CGRP's protective effects. Indeed, in patients treated with galcanezumab, it caused constipation as a side effect [187]. Certainly, since the mAbs fremanezumab, galcanezumab, and eptinezumab cannot distinguish between the α -CGRP and β -CGRP isoforms [188–190], peripheral adverse events are expected.

With respect to potential cardiovascular side effects, this concern is more than justified in view that migraine patients may be at increased cardiovascular risk [84]. Indeed, current clinical trials indicate that the four mAbs failed to produce cardiovascular adverse effects in the short-term (<6 months), but some cardio- and cerebrovascular events such as atherosclerosis, right thalamic infarction, or ischemia could favor in the long term [8,11]. More recently, in a RCT, the long-term safety (52 weeks) and efficacy of galcanezumab was evaluated. Apart from the fact that patients reported that they were extremely satisfied over prior prophylactic treatments, an improvement from baseline anxiety, depression and sleep quality was also recorded [191]; furthermore, the most common adverse events were an injection-site-reaction (26–33% of all patients) and a low incidence of cardiovascular and cerebrovascular adverse related-events [192]. Accordingly, additional long-term studies are still required to evaluate the cardiovascular safety of mAbs, particularly in: (i) women during pregnancy or after menopause (who may have a higher cardiovascular risk during CGRP/receptor blockade than men); (ii) patients with cardiovascular diseases; and (iii) animal models mimicking these pathologies [8,30].

Moreover, it is reasonable to consider that the long-term treatment with the antimigraine mAbs may induce the production of autoantibodies against these mAbs which would, in turn, neutralize their antimigraine actions. Certainly, this autoantibodies' production could represent the most significant adverse effect provoked by these mAbs [8,30]. In keeping with this view, long-term treatment with erenumab has been shown to induce autoantibodies in 10% of the patients treated [193,194].

4.1.5. Current knowledge on the safety and efficacy of the use of gepants and mAbs

As previously reviewed in detail [4,8,11,30,188], several lines of evidence have shown and confirmed that the short term use

of gepants and mAbs in the acute and prophylactic treatment of migraine is effective, well-tolerated, devoid of important side effects and safe (particularly in terms of hepatotoxicity). However, as pointed out earlier, long-term clinical studies are still required to explore and document their gastrointestinal and cardiovascular safety (among other potential side effects). In this respect, some preclinical studies imply that CGRPergic blockade may induce, among other side effects, constipation, alterations in homeostatic functions of the pituitary hormones, or attenuation of wound healing [195]. Although some of these effects have not yet been reported in humans so far, it appears that the advantages of CGRPergic blockade in migraineurs outweigh the disadvantages.

4.1.6. Is the anti-CGRPergic pharmacotherapy really effective?

Admittedly, the current antimigraine therapy with the gepants and mAbs is not effective in all migraineurs [30]. Furthermore, when comparing the therapeutic gain (*i.e.*, the effect of the active drug minus placebo) of sumatriptan (and also of the second generation triptans) with that of the novel anti-CGRPergic drugs, it would seem that the antimigraine action of the latter agents is not much improved (see Figure 3); nevertheless, the gepants have a better security profile in view that the adverse events elicited are similar to those produced in patients treated with placebo [196–199].

In the case of mAbs, as extensively discussed and illustrated by Tfelt-Hansen *et al.* [207], the methods used in the RCTs to evaluate the effectivity of the mAbs against episodic migraine have some shortcomings limiting the interpretation of the effectivity of these molecules. Indeed, the median persisting migraine monthly days (MMD) (from seven RCTs) were 55% (range 45–64%) in patients treated with mAbs, whereas in the case of a placebo, the median persisting MMD were 75% (range 69–78%) [207]. Taken together, these data imply that the net therapeutic gain is nearly 20% after three months; *i.e.*, a patient with 10 MMD, after treatment, the MMD would be reduced to 8. Nevertheless, we must keep in mind that some patients could be non-responders.

One interesting point with the use of mAbs is the probability that chronic refractory migraine could turn into episodic migraine. Indeed, in a couple of recent reports in patients diagnosed with chronic refractory migraine and treated with erenumab [187,208], the monthly headache days (MHD) were reduced from 26.5 days (20–30 days, interquartile range) to less than 14 days after four months of treatment, in at least two-thirds of treated patients. Furthermore, in 25% of the responsive patients to erenumab, the MHD was < 3 days. The contrasting effectivity in this set of patients compared to the previous RCT discussed by Tfelt-Hansen *et al.* [207] may imply that some patients are better candidates to receive these expensive agents [198].

Besides, these data imply that, in some patients, several other molecules may also play a role in migraine pathophysiology. As described elsewhere in this review, among these candidate molecules, PACAP, OT, and orexins may also be involved. Thus, a multi-target drug approach could be the

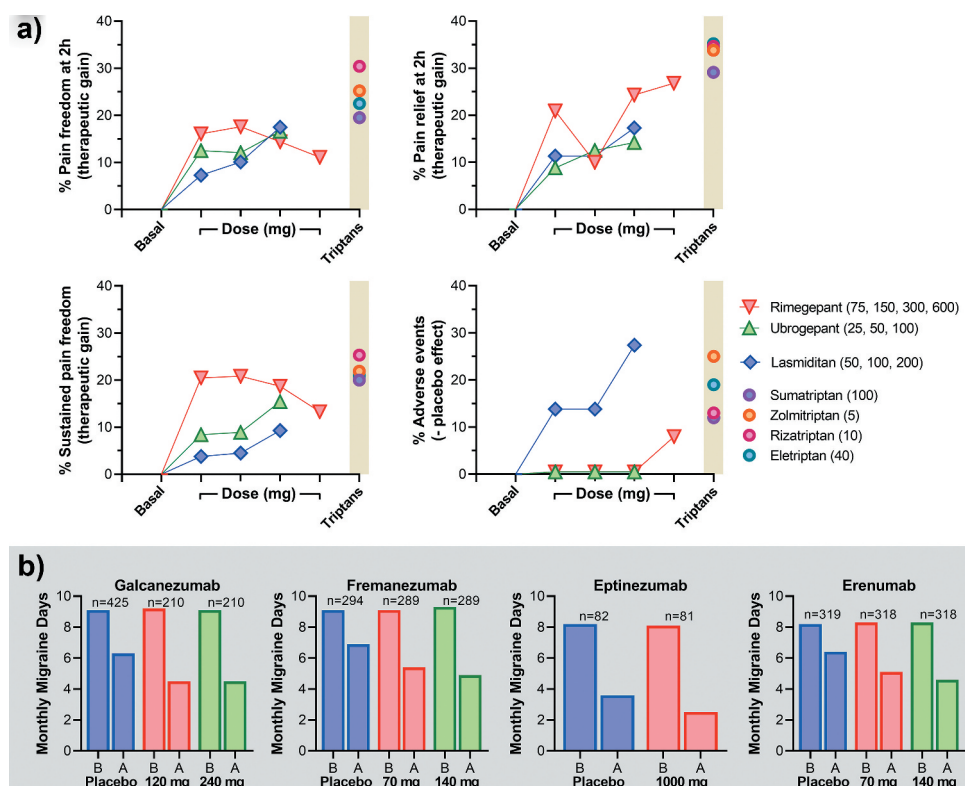


Figure 3. An overview from RCTs on the effect of the new anti-CGRP agents against migraine. (a) Therapeutic and adverse events (expressed as a response percentage) of primary (pain freedom at 2 h) and secondary (pain relief at 2 h or sustained pain freedom) endpoints obtained with different acute oral antimigraine drugs. Note that, although the therapeutic gain of gepants is lower than that of the triptans, the induction of adverse events elicited by these drugs is much lower. In contrast, although lasmiditan produced no relevant vascular effects, this drug causes mainly CNS adverse events (see Table 3 for details). The data of the different triptans were inserted for comparison. (b) Therapeutic effect of current prophylactic treatments elicited by mAbs. In this case, the effect of the different molecules is expressed as monthly migraine days with or without (placebo) treatment; note the high placebo effect observed in the RCT designed to test the clinical utility of mAbs. In this panel, the doses of mAbs agents are expressed in mg and the n of patients is indicated above the graph bars. For details see [11,37,188,197,200–202,203,204,205,206,207]

next step in the development of successful antimigraine therapies [209].

4.2. The PACAP, OT and orexin systems as potential targets to treat migraine

4.2.1. Agents against PACAP and/or the PAC₁ receptor – current RCT

As discussed above, PACAP and its receptor (*i.e.*, PAC₁) have been implicated in migraine pathophysiology; consequently, molecules to target this system have been developed. Currently, at least two mAbs targeting PACAP (ALD1910) or PAC₁ receptors (AMG 301) are in the pipeline and were developed to be used as prophylactic treatments [210]. In this sense, although AMGEN® produced and started a Phase II RCT [133] designed to evaluate the safety and efficacy of AMG 301 in patients with chronic or episodic migraine, this mAb failed to show a significant therapeutic antimigraine effect [133,211]; moreover, fatigue and constipation (>3%) were reported as the main side effect induced by this mAb [133].

The above results contrast with the primary studies suggesting that the PAC₁ receptor plays a role in the

sensitization of the TCC [135]. However, we must keep in mind that mAbs are agents with a high molecular weight and, consequently, a limited diffusion to the CNS is achieved. In agreement with these findings, the PAC₁ receptor is not expressed in the vascular tissue [212], but it still can be found in several areas of the CNS associated with migraine [213] and in the sphenopalatine ganglion [120], a structure related to the pathophysiology of other primary headaches (*i.e.*, cluster headaches and trigeminal autonomic cephalalgias (TACs)) [2]. In addition, PACAP is closely related to VIP [116], and both neuropeptides are expressed in the parasympathetic fibers innervating craniovascular structures [214]. Indeed, infusion of VIP in healthy volunteers induced delayed headache and extracranial vasodilatation, probably mediated by the cranial parasympathetic system [215]. From these data, the possibility that AMG 301 will be repurposed to test in other RCT designed to measure the efficacy in cluster headache or TACs remains open. In any case, coupled to the clinical studies showing that infusion of PACAP-38 produces vasodilatation [130] and a delayed-type migraine like-attack [128], these data imply that the PAC₁ receptor could be relevant at central, rather than

peripheral, structures in migraine [135]. Furthermore, in view that VIP induces vasodilatation of porcine basilar arteries by activation of VPAC₁ and VPAC₂ receptors [216], we could infer that PACAP-38 could be mediating vasodilatation by activation of these receptors.

In the case of ALD1910, this mAb developed by Alder® (recently acquired by Lundbeck®) targets the PACAP-38 (pK_D: 10.66) and PACAP-27 (pK_D: 10.23) [217] and it is under clinical testing evaluating safety, tolerability, and pharmacokinetics [218]. As reported by Moldovan [217], this mAb effectively blocks the PACAP signaling mediated by activation of PAC₁, VPAC₁, and VPAC₂ receptors. Moreover, in an *in vivo* rat model of parasympathetic activation by umbellulone, the pretreatment with ALD1910 (given 24 h before umbellulone) inhibited the lacrimation and flushing induced by umbellulone. Future RCTs are expected to fully uncover the role of blocking PACAP signaling as a new approach in treating primary headaches.

4.2.2. The oxytocinergic system in pain transmission – the intranasal route to reach the trigeminovascular system

As mentioned above, OT seems to play a crucial role as an endogenous molecule modulating nociception at peripheral, spinal, and supraspinal levels [144–146,153–155]. Significantly, some studies in humans showed that peripheral [158], intrathecal [157] or intracerebroventricular [219] injections of this peptide result in an analgesic action. In all cases, this analgesic action seems to be mainly mediated by OT receptor activation, but the role of V_{1A}R cannot be categorically excluded [150]. Despite its exciting analgesic properties, OT is not used in pain treatment because it cannot efficiently penetrate the CNS. Besides its relatively high molecular weight (~ 1 kDa), this peptide has a low oral absorption and it is rapidly metabolized after intravenous (3–5 min) [220,221] or intracerebroventricular (~ 20 min) [222] administration.

Currently, only one nonpeptide liposoluble OT receptor agonist that is capable of crossing the BBB has been reported, namely, LIT-001 [223]. Although this agonist has a preferential affinity for the OT receptor (K_i: 6.7), it is not that selective as it still displays some affinity for the V_{1A}R (pK_i: 5.9). In any case, in a recent study using a model of carrageenan-induced inflammatory pain, Hilfiger *et al.* [224] showed that intraperitoneal administration of LIT-001 induces a long-lasting reduction in hyperalgesia. In this context, future studies in rodents, analyzing the effect of intravenous LIT-001 in the trigeminovascular system, will shed further light on the clinical feasibility to target this system by using nonpeptide OT receptor agonists.

Certainly, electrophysiological, behavioral and molecular data support the notion that the oxytocinergic system plays a pivotal role as an endogenous hypothalamic molecule modulating the trigeminovascular nociception [162–167]. Accordingly, it has been shown that the OT receptor is expressed in the trigeminal ganglion [167], particularly in CGRP-positive neurons [225]. At this point, although OT exerts antinociception at different levels of the trigeminovascular system, an array of potential mechanisms to exert analgesia are recruited [141,150,151]. Furthermore, it is noteworthy that

OT plays a neuromodulatory role at the cortical, subcortical and brainstem levels by modulating not only the sensory, but also the emotional aspects of pain processing [139].

Regarding the emotional aspects, and beyond migraine (or pain), the role of central OT signaling in humans has been widely studied using an intranasal OT administration to overpass the BBB and reach the CNS (for references see [226]). Most interestingly, and focusing on the safety issues of intranasal OT administration, the global clinical experience on the use of intranasal OT has reported that:

- (i) After analyzing 38 RCTs, MacDonalds *et al.* [227] found that in a population of 1529 adults (79% male, 21% female), the short-term use of (mainly single) intranasal OT given in a dose-ranging 18–40 international units (UI; ~ 36–80 µg), produces no detectable subjective changes or relevant side-effects (as compared vs. placebo).
- (ii) When using a double-blind RCT to examine the side-effects induced by intranasal OT in a sample of 100 healthy children (8–12 years old; 46 boys and 54 girls), Verhess *et al.* [228] found that the side-effects induced after a single intranasal OT dose (12 IU [~ 24 µg]/40 kg) were not more frequent than those observed after placebo.

Notwithstanding, no RCT has yet been performed to analyze the case of long-term use of intranasal OT. In this sense, since intranasal OT administration has long been used as a route for central delivery, Wang *et al.* [161] reported that intranasal OT relieves headache; this study was performed in 112 patients diagnosed with tension-type headache and migraines. In contrast, in a pilot double-blind placebo-controlled study (NCT00963040 and NCT01839149), Tzabasis *et al.* [160] found no difference in pain reduction vs. the placebo group; in this case, OT (32 UI) was given at the onset of the headache, and pain relief was evaluated 2-h after dosing intranasal OT. It is interesting to note that, apart from the use of a single dose of OT, the drug was given when the migraine headache had been established, and a high placebo response was achieved in the population tested (26%). Interestingly, in these patients, 2-h after OT, an apparent reduction in photophobia (21% vs. 9% in placebo), phonophobia (16% vs. 2% in placebo), and nausea (27% vs. 17% in placebo) was achieved.

In the same report [160], but analyzing the effect of daily (for 28 days) recurrent intranasal OT administration in patients with chronic (>14 MMD) or high-frequency episodic migraine (>8 < 14 MMD), the authors found that the total MMD were decreased in severity and frequency. These data may imply that OT has an impact on the frequency by modulation of neuronal activity, probably by enhancing endogenous analgesic systems. For example, a recent report showed that OT microinjection at the insular cortex seems to recruit the GABAergic circuits by promoting the activation of descending antinociceptive mechanisms [229].

Beyond the questions on whether intranasal OT administration reaches the CNS at therapeutic concentrations

[226,230,231], it has been shown in rodents, using radiolabelled ^{125}I -OT, that intranasal OT reaches critical structures related with the nociceptive processing at the spinal trigeminal nucleus and the trigeminal ganglion. Furthermore, also structures coupled not only with pain integration, but also with the emotional aspects of pain are radiolabelled (*i.e.*, medulla, hypothalamus, amygdala, and prefrontal cortex) [160]. These data support the notion that after intranasal OT, areas related to migraine headaches could be modulated by this neuropeptide.

Despite the potential use of intranasal OT as an antimigraine agent, no current clinical trials are active. In general, since 2013, only seven clinical trials have been registered with the intention to evaluate if OT by intranasal route (32–48 IU) has any effect on different pain conditions (excluding migraine). Currently, three studies have been completed (NCT01886014, NCT03060031, and NCT00919802), two are active (NCT03878589 and NCT04218409), and the status of the two remaining trials (NCT03122067 and NCT02550093) is unknown.

4.2.3. Ligands interacting with the orexinergic system – the failure of the DORA ligands in RCT

As extensively reviewed [231,232], the orexinergic system is composed by the neuropeptides orexin-A and orexin-B, and its receptors, the OX_1 and OX_2 , both preferentially coupled to G_q proteins. Although the activation of this system at the posterior hypothalamus was initially related to the sleep/wake process and feeding behavior, this system is currently involved in the reward systems and as a relevant modulator of the hypothalamic-pituitary-adrenal axis. Moreover, since orexinergic fibers project to the spinal dorsal horn, and both orexins are expressed at this level [233,234], a role for nociception modulation has been hypothesized. Indeed, since intrathecal and intracerebroventricular administration of orexin-A produces antinociception, the relevance of this peptide as an analgesic compound has been suggested [235–237].

Considering that the posterior hypothalamus is involved in migraine pathophysiology [238,239], and that both orexins can be synthesized at this level, a role for the orexinergic system was suggested [240]. In this respect, a seminal study by Bartsch *et al.* [174] showed that microinjection of orexin-A at the posterior hypothalamus inhibited the dural-evoked nociceptive responses at TCC, whereas orexin-B microinjection facilitated the TCC nociception. Furthermore, intravenous administration of the dual orexin receptor antagonist 12 (DORA-12) resulted in inhibition of the dural evoked neuronal responses at TCC [241]. These results pave the way for the above compounds in migraine treatment.

Unfortunately, in an RCT (NCT01513291) using patients diagnosed with episodic migraine, the treatment with filorexant (a DORA compound developed for insomnia) for three months showed no antimigraine effect. Besides, in this Phase II study, no relevant adverse events occurred. However, as pointed out by Strother *et al.* [10], the negative results in this RCT may have been caused by the nighttime dosing and short half-life (3–5 h) of this molecule. Indeed, this schedule was chosen to avoid somnolence, and

probably a better strategy would have been to analyze the antimigraine effects of DORAs in a subpopulation of migraine patients with insomnia. In any case, no active clinical trials are currently running.

At this point, we must recapitulate some critical issues on the role of orexins in migraine pathophysiology, briefly:

- (i) Using intravital microscopy and intravenous administration of orexin-A or orexin-B, it has been shown that the neurogenic dural vasodilatation can be inhibited only by orexin-A via prejunctional activation of peripheral OX_1 receptors [242].
- (ii) Similarly, when the dural-evoked neuronal responses at TCC were analyzed, orexin-A, but not orexin-B, inhibits selectively the neuronal firing of A-fibers but not C-fibers. This neuronal inhibition was associated to OX_1 receptor activation, since SB-334,867 (an OX_1 receptor antagonist) blocked the orexin-A-induced antinociception [173].
- (iii) More recently, in a rat behavioral migraine model using nitroglycerin (NTG), Kooshki *et al.* [243] showed that microinjection of orexin-A in the ventrolateral PAG not only attenuated the NTG-induced behavioral nociception, but also the NTG-induced CGRP overexpression in the trigeminal caudal nucleus. This antinociception induced by orexin-A was blocked by SB-334,867.
- (iv) Together, these results may imply that a selective OX_1 receptor agonist is a better candidate to block the trigeminovascular nociception, rather than DORAs. Clearly, the further development of selective OX_1 receptor agonists may unravel the role of this receptor in migraine pathophysiology.

5. Expert opinion

Migraine is a complex neurovascular disorder in which the trigeminovascular system plays a fundamental role both in its pathophysiology and in the development of anti-migraine pharmacotherapies. Moreover, it is currently accepted the importance of: (i) peripheral vasodilatation (*i.e.*, in intracranial blood vessels and meninges) mainly driven by CGRP release from the third branch (ophthalmic) of the trigeminal nerve; (ii) a nociceptive sensitization at the spinal trigeminal level facilitated by CGRP; and (iii) the role of the brainstem and hypothalamic structures.

Despite the debate about how antimigraine drugs work, evidence from basic and clinical reports from the last fifteen years suggests that central actions are not necessary to exert an antimigraine effect. Indeed, the novel therapy with mAbs strongly supports this idea since these molecules cannot cross the BBB due to their high molecular weights. Moreover, since gepants and mAbs cause adverse events in the same proportion as placebo, the actions of these new molecules are restricted to the periphery. In agreement with this view, Hostetler *et al.* [23] used radiolabelled [^{11}C]MK-4232 to displace telcagepant from its binding sites and showed in humans that after an intravenous

infusion of this gepant, an irrelevant concentration of telcagepant was found in the CNS. Similarly, Nosedá *et al.* [24] used a fluorescent-labeled fremanezumab to demonstrate that only structures outside the BBB related with migraine pathophysiology (*i.e.*, dural tissue, trigeminal and C2 dorsal root ganglion, and the parasympathetic sphenopalatine ganglion) were labeled, and no detectable levels of this mAb were found in the CNS.

Despite the evidence supporting that the therapeutic actions of gepants and mAbs are restricted to the periphery, the therapeutic gain exhibited in an RCT in migraine patients with these treatments is rather low (see Figure 3). Moreover, in the case of acute pharmacotherapy, the gepants have a lower therapeutic gain (7–12%) as compared to sumatriptan (21%), eletriptan (26%), or rizatriptan (32%) (Figure 3). Similarly, as shown in the Figure 3(b), after mAb treatment and considering the therapeutic gain, the MMD seems to diminish 20% (in the best case) in patients diagnosed with episodic migraine. However, some data suggest that these mAbs could be a better option for some patients with chronic migraine, where these molecules convert the chronic migraine to episodic migraine after three months of treatment [187,191].

Overall, these data imply that the development of a ‘magic antimigraine bullet’ seems not to work efficiently. Certainly, the novel antimigraine pharmacotherapies are designed to specifically target the CGRPergic system in the periphery of the trigeminovascular system; nevertheless, it is noteworthy that other parts in the CNS and parasympathetic system are also involved. In this respect, we consider that triptans and ergots are more effective to palliate migraine since these molecules target an array of receptors (*e.g.*, 5-HT_{1B/1D}, $\alpha_{2A/2C}$ -adrenoceptors) which are expressed in the cranial vasculature (exerting vasoconstriction), at the prejunctional perivascular level (inhibiting CGRP release and other vasoactive substances like PACAP), and at the trigeminal and TCC levels modulating (inhibiting) the peripheral nociceptive input; hence these pleiotropic actions may boost the therapeutic actions of these drugs. However, being nonspecific, they cause an array of adverse events [11]. Consequently, the search for new targets against migraine is relevant.

Since neuroimaging studies point out the role of the hypothalamus in migraine [20,96], and several symptoms before, during, and after a migraine attack could be associated with a hypothalamic dysfunction [98], the role of some hypothalamic neuropeptides has emerged [73,95]. In general, these neuropeptides may exert a series of neuromodulatory actions at the nervous systems, and preclinical data suggest that the PACAPergic, oxytocinergic, and orexinergic systems could be a druggable target for antimigraine treatment. In this context, although the PACAPergic system emerged as a promising target for the prophylactic treatment of episodic migraine, and the RCT using AMG 301 (mAb against the PAC₁) recorded a low incidence of adverse events, this mAb failed to show a therapeutic effect [133,211]. At this point, one plausible explanation on this lack of therapeutic effect could be that the PAC₁ receptor is not preferentially expressed in craniovascular tissue or peripheral trigeminal neurons. Indeed, the PAC₁ receptor can be found in the parasympathetic sphenopalatine ganglion and, consequently, this mAb could be repurposed to

be analyzed in other primary headaches related to autonomic cephalalgias. Besides, an RCT evaluating the effect of ALD 1910 against PACAP-38 and PACAP-27 is still active [218]; hence, an antimigraine effect will probably be achieved since this mAb blocked the PACAP signaling mediated not only by PAC₁, but also by VPAC₁ and VPAC₂ receptors. As a result, the role of VIP as a neuropeptide modulating the trigeminovascular system has been proposed.

In the case of OT, this neuropeptide exerts an analgesic effect when given at spinal or supraspinal levels. Indeed, OT inhibits the trigeminovascular nociception at peripheral and TCC levels; nonetheless, the main issue is that no selective non-peptide OT receptor agonists have been described thus far. Interestingly, in the psychiatric research field, intranasal OT (which overpasses the BBB) has been used to investigate its effects in the CNS; nevertheless, no consistent antimigraine effects have been obtained.

On the other hand, the role of orexins modulating the TCC activity by peripheral or hypothalamic descending mechanisms has been demonstrated; however, an RCT testing the potential antimigraine action of filorexant (a DORA molecule) showed no antimigraine effect.

In conclusion, although PACAP, OT, and orexins appear to play a role in migraine pathophysiology in basic studies, the translation to the clinic has been more challenging than initially expected. In this sense, it cannot be overemphasized that the antimigraine clinical experience with the triptans represents about 30 years. In contrast, the antimigraine clinical experience with the ditans, gepants and mAbs is more limited (around 6 years, including short-term RTCs), but it is certainly encouraging, exciting and very promising. Nevertheless, Phase IV RCTs with these novel antimigraine drugs will shed further light on their efficacy and safety.

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